

3(a)	$E_{(P)} = \frac{1}{2}kx^2$ or $E_{(P)} = \frac{1}{2}Fx$ and $F = kx$	C1
	$0.048 = \frac{1}{2} \times k \times (2.1 \times 10^{-2})^2$ $k = 220 \text{ N m}^{-1}$	A1
3(b)	$E_{(K)} = \frac{1}{2}mv^2$	C1
	$0.048 = \frac{1}{2} \times 7.5 \times 10^{-3} \times v^2$ $v = 3.6 \text{ m s}^{-1}$	A1
3(c)(i)	$(\Delta)E = mg(\Delta)h$	C1
	$0.039 = 7.5 \times 10^{-3} \times 9.81 \times (\Delta)h$ $\Delta h = 0.53 \text{ m}$	A1
3(c)(ii)	$F \times 0.53 = 0.048 - 0.039$	C1
	$F = 0.02 \text{ N}$	A1
3(d)	sketch: curved line from the origin	M1
	curved line has increasing gradient	A1